

Topic 11: Second-Order Linear Homogeneous Equations & The 3 Characteristic Cases

Professor Leonard Master Series · Lessons 37 to 39

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Characteristic Bridge

Professor Leonard reveals the core mystery of 2nd-order constant coefficient ODEs:

"Look at $ay'' + by' + cy = 0$. What function on Earth has the property that its derivatives are multiples of itself, so they can add up to zero? The Exponential Function e^{rx} ! If you test $y = e^{rx}$, the derivatives bring down powers of r : $y' = re^{rx}$ and $y'' = r^2e^{rx}$. Factoring out $e^{rx} \neq 0$ turns calculus into high school algebra: $ar^2 + br + c = 0$! Every 2nd-order ODE is governed entirely by the roots of that quadratic polynomial!"

2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 11.1: Case 1 — Distinct Real Roots & Initial Value Problem

Statement: Solve the IVP:

$$y'' - 5y' + 6y = 0, \quad y(0) = 2, \quad y'(0) = 5$$

Step-by-Step Whiteboard Solution:

- **Step 1: Write the Characteristic (Auxiliary) Equation:**

$$r^2 - 5r + 6 = 0$$

- **Step 2: Factor and solve for roots:**

$$(r - 2)(r - 3) = 0 \implies r_1 = 2, \quad r_2 = 3$$

- **Step 3: Write the General Solution:** Since roots are real and distinct:

$$y(x) = c_1e^{2x} + c_2e^{3x}$$

- **Step 4: Compute $y'(x)$:**

$$y'(x) = 2c_1e^{2x} + 3c_2e^{3x}$$

- **Step 5: Apply Initial Conditions at $x = 0$:**

$$y(0) = c_1 + c_2 = 2$$

$$y'(0) = 2c_1 + 3c_2 = 5$$

- **Step 6: Solve the 2×2 algebraic system:** From (1), $c_1 = 2 - c_2$. Substitute into (2):

$$2(2 - c_2) + 3c_2 = 5 \implies 4 - 2c_2 + 3c_2 = 5 \implies c_2 = 1$$

$$c_1 = 2 - 1 = 1$$

- **Step 7: Final Unique Solution:**

$$y(x) = e^{2x} + e^{3x}$$

Problem 11.2: Case 2 — Repeated Real Root & The Magic Factor of x

Statement: Solve the IVP:

$$y'' + 6y' + 9y = 0, \quad y(0) = 1, \quad y'(0) = -1$$

Step-by-Step Whiteboard Solution:

- **Step 1: Characteristic Equation:**

$$r^2 + 6r + 9 = 0 \implies (r + 3)^2 = 0 \implies r_1 = r_2 = -3$$

- **Step 2: Professor Leonard's Epiphany on Linear Independence:** We found only one exponential solution $y_1(x) = e^{-3x}$. But a 2nd-order ODE requires TWO linearly independent solutions! Using Reduction of Order: $y_2(x) = xe^{-3x}$. Fundamental Solution Set: e^{-3x}, xe^{-3x} .
- **Step 3: General Solution:**

$$y(x) = c_1 e^{-3x} + c_2 x e^{-3x} = e^{-3x}(c_1 + c_2 x)$$

- **Step 4: Compute $y'(x)$ using Product Rule:**

$$y'(x) = -3c_1 e^{-3x} + c_2 [e^{-3x} - 3x e^{-3x}] = e^{-3x} [(-3c_1 + c_2) - 3c_2 x]$$

- **Step 5: Apply Initial Conditions:**

$$y(0) = c_1 = 1$$

$$y'(0) = -3c_1 + c_2 = -1 \implies -3(1) + c_2 = -1 \implies c_2 = 2$$

- **Step 6: Final Solution:**

$$y(x) = e^{-3x}(1 + 2x)$$

Problem 11.3: Case 3 — Complex Conjugate Roots & Euler's Formula

Statement: Solve the IVP:

$$y'' + 4y' + 13y = 0, \quad y(0) = 3, \quad y'(0) = -6$$

Step-by-Step Whiteboard Solution:

- **Step 1: Characteristic Equation:**

$$r^2 + 4r + 13 = 0$$

- **Step 2: Quadratic Formula:**

$$r = \frac{-4 \pm \sqrt{16 - 52}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = \frac{-4 \pm 6i}{2} = -2 \pm 3i$$

Real part $\alpha = -2$, Imaginary part $\beta = 3$.

- **Step 3: General Solution via Euler's Identity** ($e^{i\theta} = \cos \theta + i \sin \theta$):

$$y(x) = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x) = e^{-2x} (c_1 \cos 3x + c_2 \sin 3x)$$

- **Step 4: Compute $y'(x)$ using Product and Chain Rules:**

$$y'(x) = -2e^{-2x} (c_1 \cos 3x + c_2 \sin 3x) + e^{-2x} (-3c_1 \sin 3x + 3c_2 \cos 3x)$$

- **Step 5: Apply Initial Conditions:**

$$y(0) = e^0 (c_1 \cos 0 + c_2 \sin 0) = c_1 = 3$$

$$y'(0) = -2(3)(1) + 3c_2(1) = -6 + 3c_2 = -6 \implies 3c_2 = 0 \implies c_2 = 0$$

- **Step 6: Final Solution (Pure Damped Cosine):**

$$y(x) = 3e^{-2x} \cos 3x$$

Problem 11.4: Reduction of Order (Deriving y_2 from y_1)

Statement: Given that $y_1(x) = x^2$ is a solution to $x^2 y'' - 3xy' + 4y = 0$ on $x > 0$, find the second linearly independent solution $y_2(x)$.

Step-by-Step Whiteboard Solution:

- **Step 1: Standard form:**

$$y'' - \frac{3}{x}y' + \frac{4}{x^2}y = 0 \implies P(x) = -\frac{3}{x}$$

- **Step 2: Apply Professor Leonard's Reduction of Order Formula:**

$$y_2(x) = y_1(x) \int \frac{e^{-\int P(x)dx}}{[y_1(x)]^2} dx$$

- **Step 3: Evaluate Numerator:**

$$e^{-\int -\frac{3}{x} dx} = e^{3 \ln x} = x^3$$

- **Step 4: Integrate:**

$$y_2(x) = x^2 \int \frac{x^3}{(x^2)^2} dx = x^2 \int \frac{x^3}{x^4} dx = x^2 \int \frac{1}{x} dx = x^2 \ln x$$

- **Conclusion:** General solution is $y(x) = c_1 x^2 + c_2 x^2 \ln x$.

Problem 11.5: The Wronskian & Abel's Theorem

Statement: For $xy'' + 2y' + xe^x y = 0$, compute the Wronskian $W(x)$ up to an arbitrary constant without solving the differential equation!

Step-by-Step Whiteboard Solution:

- **Step 1: Standard Form:**

$$y'' + \frac{2}{x}y' + e^x y = 0 \implies P(x) = \frac{2}{x}$$

- **Step 2: Apply Abel's Formula:**

$$W(x) = C e^{-\int P(x)dx} = C e^{-\int \frac{2}{x} dx} = C e^{-2 \ln x} = C x^{-2} = \frac{C}{x^2}$$

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Forgetting the x -factor for repeated roots:** Writing $y = c_1 e^{rx} + c_2 e^{rx} = (c_1 + c_2)e^{rx} = C e^{rx}$ leaves you with only ONE constant!
- **Trap 2: Imaginary unit i in solution:** Do NOT include i in your final solution for Case 3! Linear combinations of $e^{(\alpha \pm i\beta)x}$ cancel i , yielding real $\cos \beta x$ and $\sin \beta x$.